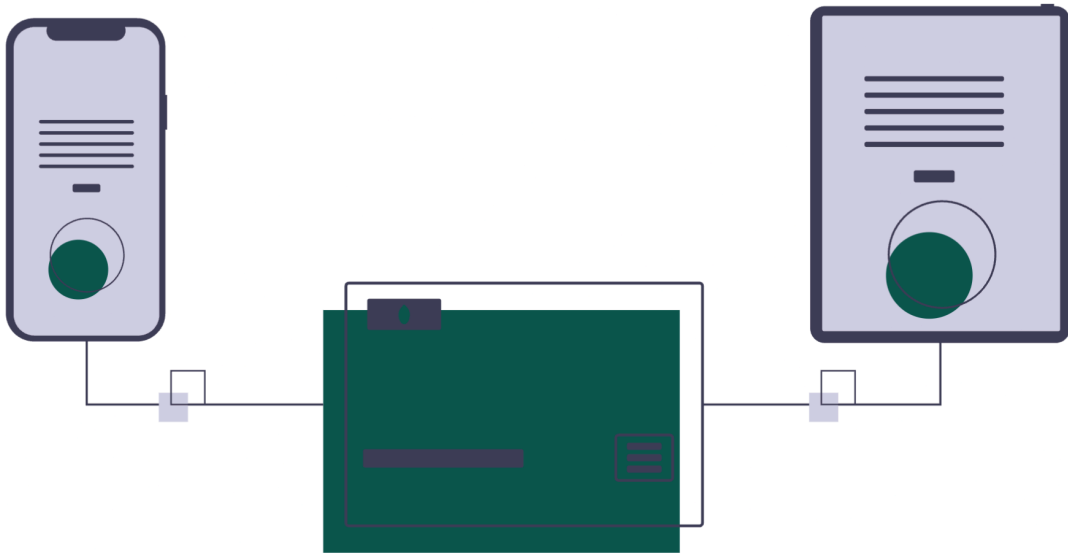


Lab Final Assignment of CSE405

Computer Networks



Submitted By:

Aklhak Hossain

2022-3-60-057

<https://akhlak.com>

Submitted To:

Md. Khalid Mahbub Khan

Senior Lecturer

Department of CSE

East West University

Objective:	3
Network Design & IP Scheme.....	3
LAN Networks (Class C):.....	3
WAN Networks (Class B: for inter-router links):.....	3
Device Configuration:.....	3
DHCP:.....	3
EIGRP Routing:.....	3
Network Architecture:.....	4
Figure: Network Topology.....	4
IP Addressing Scheme	5
Table : Network & Host IP Addresses.....	5
Hardware & Software.....	6
Experimental Results:	6
Routing Configuration:.....	6
Router 0:.....	6
EIGRP 1:.....	6
Router 1:.....	7
EIGRP 2:.....	7
Router 2:.....	7
EIGRP 3.....	8
Router 3:.....	8
EIGRP 4.....	9
Server Configuration Results:	9
Testing & Verification.....	10
Conclusion	11

Objective:

This lab final focuses on designing, setting up, and troubleshooting complex network topologies in Cisco Packet Tracer. The work is split into two parts:

1. Building a 4-router network that incorporates DHCP, EIGRP routing, and a combined DNS/Web server.
2. Constructing a fault-tolerant 4-router campus network using both linear and loop connections, without relying on any routing protocol.

Network Design & IP Scheme

A four-router network was built following a provided diagram, using both Class B and Class C IP address ranges.

LAN Networks (Class C):

- Router 0: 192.168.10.0
- Router 2: 194.168.20.0
- Router 3: 193.168.10.0

WAN Networks (Class B: for inter-router links):

- Address blocks 172.16.0.0, 173.16.0.0, 174.16.0.0, and 175.16.0.0 were assigned to the serial links between routers.

Device Configuration:

DHCP:

Individual DHCP pools were set up on the edge routers (Routers 0, 2, and 3) so that connected end devices could automatically obtain their IP address, subnet mask, default gateway, and DNS server information. A sample configuration for Router 0 is shown below:

```
Router(config)# ip dhcp pool LAN0
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.254
Router(dhcp-config)# dns-server 192.168.10.100
```

EIGRP Routing:

EIGRP was deployed across all four routers to provide end-to-end connectivity. The no auto-summary command was essential to avoid route summarization conflicts between the Class B and Class C address spaces:

```
router eigrp 1
network 192.168.10.0 0.0.0.255
network 172.16.0.0 0.0.255.255
network 174.16.0.0 0.0.255.255
exit
```

Network Architecture:

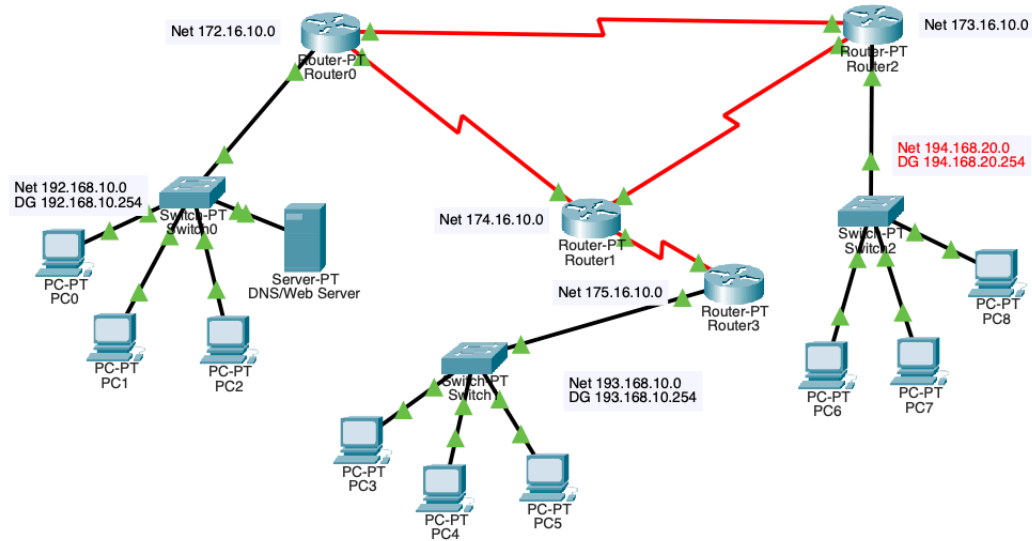


Figure: Network Topology

IP Addressing Scheme

Router	Interface	IP Address	Subnet Mask	Description
Router 0	fa0/0	192.168.10.254	255.255.255.0	LAN 1
	se2/0	172.16.10.1	255.255.0.0	Link to Router 1
	se3/0	174.16.10.2	255.255.0.0	Link to Router 2
Router 1	se2/0	173.16.10.1	255.255.0.0	Link to Router 2
	se3/0	172.16.10.2	255.255.0.0	Link to Router 1
	se7/0	175.16.10.2	255.255.0.0	Link to Router 3
Router 2	fa0/0	194.168.20.254	255.255.255.0	LAN 2
	se2/0	174.16.10.1	255.255.0.0	Link to Router 1
	se3/0	173.16.10.2	255.255.0.0	Link to Router 0
Router 3	fa0/0	193.168.10.254	255.255.255.0	LAN 3
	se2/0	175.16.10.1	255.255.0.0	Link to Router 1

Table : Network & Host IP Addresses

Hardware & Software

- **Simulator:** Cisco Packet Tracer
- **Devices:** 4 Cisco routers, 3 switches, multiple PCs, and generic server units
- **Cabling:** Serial DCE cables for all router-to-router links; Copper Straight-Through cables for connecting end devices to switches

Experimental Results:

Routing Configuration:

Static routes were applied on all four routers to ensure complete network visibility. Full interface and routing configurations for each router are provided below.

Router 0:

```
enable
config
interface fa0/0
ip address 192.168.10.254 255.255.255.0
no shut
do wr
exit
interface se2/0
ip address 172.16.10.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 174.16.10.2 255.255.0.0
no shut
do wr
exit
```

EIGRP 1:

```
enable
config
ip dhcp pool LAN0
network 192.168.10.0 255.255.255.0
default-router 192.168.10.254
dns-server 192.168.10.100
```

```
no shut
do wr
exit
```

Router 1:

```
enable
config
interface se2/0
ip address 173.16.10.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 172.16.10.2 255.255.0.0
no shut
do wr
exit
interface se7/0
ip address 175.16.10.2 255.255.0.0
no shut
do wr
exit
```

EIGRP 2:

```
router eigrp 1
network 40.10.0.0 0.255.255.255
network 142.110.0.0 0.0.255.255
network 143.110.0.0 0.0.255.255
exit
```

Router 2:

```
enable
config
interface fa0/0
ip address 194.168.20.254 255.255.255.0
no shut
do wr
exit
interface se2/0
ip address 174.16.10.1 255.255.0.0
```

```
clock rate 64000
no shut
do wr
exit
interface se3/0
ip address 173.16.10.2 255.255.0.0
no shut
do wr
exit
```

EIGRP 3

```
router eigrp 1
network 194.168.20.0 0.0.0.255
network 173.16.0.0 0.0.255.255
network 174.16.0.0 0.0.255.255
exit
enable
config
ip dhcp pool LAN2
network 194.168.20.0 255.255.255.0
default-router 194.168.20.254
dns-server 192.168.10.100
no shut
do wr
exit
```

Router 3:

```
enable
config
interface fa0/0
ip address 193.168.10.254 255.255.255.0
no shut
do wr
exit
interface se2/0
ip address 175.16.10.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
```

EIGRP 4

```
router eigrp 1
network 193.168.10.0 0.0.0.255
network 175.16.0.0 0.0.255.255
exit
enable
config
ip dhcp pool LAN3
network 193.168.10.0 255.255.255.0
default-router 193.168.10.254
dns-server 192.168.10.100
no shut
do wr
exit
```

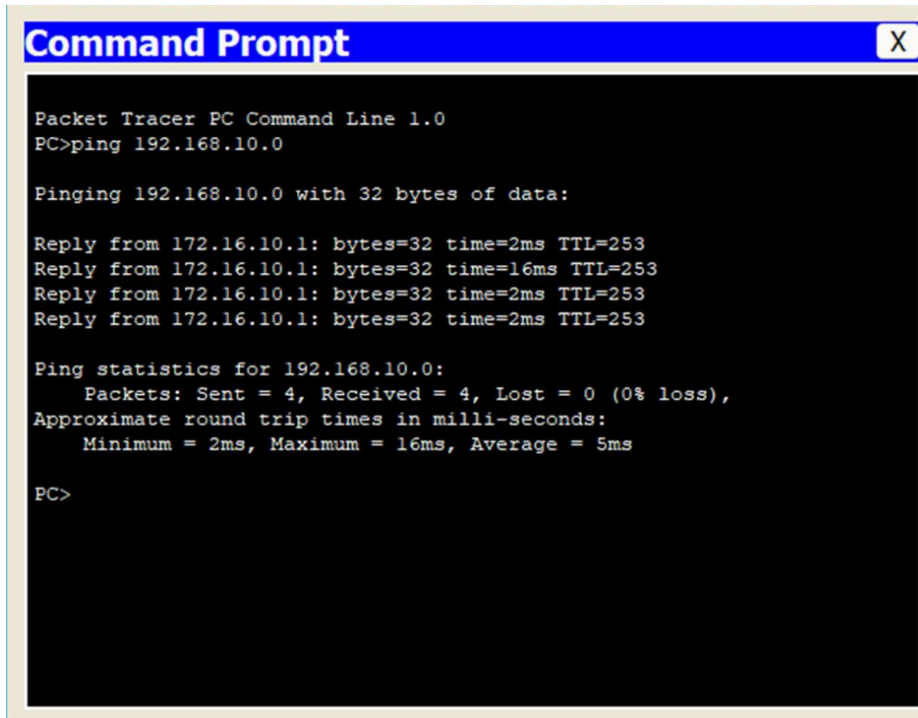
Server Configuration Results:

A Server-PT device was connected to Router 0's LAN and given the static IP address 192.168.10.100. Two services were enabled:

- **Web (HTTP):** A custom index.html page with a university theme was created, displaying the student's name and ID.
- **DNS:** A domain record was configured to resolve www.university.edu to the server's IP address (192.168.10.100).

Testing & Verification

- **Ping:** Connectivity tests from PC5 to PC8 and across other network segments all succeeded, confirming end-to-end reachability throughout the topology.



```
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.10.0

Pinging 192.168.10.0 with 32 bytes of data:

Reply from 172.16.10.1: bytes=32 time=2ms TTL=253
Reply from 172.16.10.1: bytes=32 time=16ms TTL=253
Reply from 172.16.10.1: bytes=32 time=2ms TTL=253
Reply from 172.16.10.1: bytes=32 time=2ms TTL=253

Ping statistics for 192.168.10.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 16ms, Average = 5ms

PC>
```

- **Web browsing:** Entering www.university.edu in a PC's browser correctly loaded the custom university webpage hosted on the server.



Conclusion

This lab demonstrated the full deployment of an enterprise-grade network. Key skills developed include configuring router-based DHCP for dynamic IP allocation, implementing EIGRP for multi-network routing, preventing route summarization issues, hosting DNS and web services on a local server, and designing a physically resilient mesh topology capable of surviving individual link failures.